Mathematics 629
Spring 2026
Homework Assignment 6

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Due: 3/25/2026

1. Write out the complete details of the proof of the "differentiation under the integral" Theorem 1.129 in the notes.

   Proof. i) Suppose there exists $g \in L^1(\mu)$ such that $|f(x,t)| \leq g(x)$ for all (x,t) and $\lim_{t \to t_o} f(x,t) = f(x,t_o)$. Let $f_n(x) = f(x,t_o)$ so $f_n(x) \to f(x,t)$. Since $g(x)$ dominates $f(x,t)$ and f_n converges pointwise, by DCT,
   $$\lim \int f_n d\mu = \int f(x,t_o) d\mu$$
   or equivalently $F(t_n) \to F(t_o)$. Since $t_n \to t_o$ is an arbitrary sequence then $\lim_{t \to t_o} F(t) = F(t_o)$ for all t_o.
   ii) Suppose there also exists $t \in [a,b]$ and partial derivative $\frac{\partial f}{\partial t}$ exists such that $|\frac{\partial f}{\partial t} f(x,t)| \leq g(x)$ almost everywhere. Let $d_n(x) = \frac{f(x,t+h_n - f(x,t_o)}{h_n}$ be a sequence where $h_n \to 0$. Then by definition of derivative, $d_n \to \frac{\partial f}{\partial t}$. By assumption, $|d_n| \leq |\frac{\partial f}{\partial t}| \leq g(x)$ a.e. so by DCT,
   $$\lim_{n \to \infty} \int d_n d\mu = \int \frac{\partial f}{\partial t}(x,t) d\mu.$$
   Since $\int d_n d\mu = \int \frac{f(x,t+h_n) - f(x,t_o)}{h_n} = \frac{F(t) - F(t_o)}{h_n}$ then $F'(t) \to \int \frac{\partial f}{\partial t}(x,t) d\mu$. $\square$

2. Prove that $\int_{(0,\infty)} x^n e^{-x}\, dx = n!$ by differentiating $\int_0^\infty e^{-tx}\, dt = 1/x$.

   Proof. Starting with $\int_0^\infty e^{-tx} dt = \frac{1}{x}$, differentiate both sides such that $\frac{d^n}{dx^n} \frac{1}{x} = \frac{(-1)^n n!}{x^{m+1}}$ and $\frac{d^n}{dx^n} \int_0^e {}^{-tx} dt = \int_0^\infty \frac{d^n}{dx^n}(e^{-tx}) dt = \int_0^\infty (-1)^n t^n e^{-tx} dt$. Then $\int_0^\infty t^n e^{-tx} dt = \frac{n!}{x^{n+1}}$ so when $x = 1$, $\int_0^\infty t^n e^{-tx} dt = n!$. If $x \in [a,b]$ then $t^n e^{-t} \leq t^n e^{-tx} \leq t^n e^{-ta}$so the necessary conditions of thoerem 1.129 are satisfied and the differentiation under the integral sign is valid. $\square$

3. Prove that if f_n is a sequence of decreasing nonnegative measurable functions with $\int f_1\, d\mu < \infty$, then
   $$\int \lim_{n \to \infty} f_n\, d\mu = \lim_{n \to \infty} \int f_n\, d\mu.$$

Proof. Since $\int f_1 d\mu < \infty$ then $f_1 \in L^1(\mu)$ and f_1 dominates f_n as a sequence of decreasing nonnegative measurable functions. Furthermore, $\lim_{n\to\infty} f_n = 0$ since f_n is bounded below by 0 and decreasing. Thus DCT shows $\int \lim_{n\to\infty} f_n d\mu = \lim_{n\to\infty} \int f_n d\mu$. □

4. Let $y_1, \ldots, y_M$ be distinct points in $\mathbb{R}$, and $\beta_1, \ldots, \beta_M \in \mathbb{R}$. Let

$$f(x) = \prod_{j=1}^{M} |x - y_j|^{-\beta_j}.$$

Find a necessary and sufficient condition on $(\beta_1, \ldots, \beta_M)$ for f to belong to $L^1(\mathbb{R}, m)$.

Proof. The necessary and sufficient condition for $f \in L^1(\mathbb{R}, m)$ is that $\beta_j < 1$ for all $j \in \{1, 2, \cdots M\}$ and $\sum_{j=1}^{M} \beta_j < 1$ also. $\beta_j < 1$ is necessary to prevent $f(x)$ from diverging near y_j but this isn't sufficient because a small β_j like $\beta_j = \overline{1/2M}$ would cause $\int f dm$ to diverge. Enforcing $\sum_{j=1}^{M} \beta_j < 1$ ensures $\int f dm$ is finite since $f(x) = |x - y_j|^{\beta_{product} < 1}$ whose integral does converge. □

5. Define $g \colon \mathbb{R} \to \mathbb{R}$ by

$$g(x) = \begin{cases} |x|^{-1/3}, & |x| < 1, \\ 0, & |x| > 1. \end{cases}$$

Let $\{r_k\}_{k=0}^{\infty}$ be an enumeration of all points in $[0,1] \cap \mathbb{Q}$. Let

$$f(x) = \mathbf{1}_{[0,1]} \sum_{k \in \mathbb{N}} 2^{-k} g(x - r_k).$$

(i) Show that $f \in L^1([0,1])$.

Proof. □

(ii) Show that f is unbounded on every open set (and remains so after modification on a set of measure zero). Show that f is discontinuous at every point in $[0,1]$ and remains so after modification on a set of measure zero.

Proof. □

(iii) Let $0 < \varepsilon < 1$. Construct a compact subset K_ε of $[0,1]$ such that $m(K_\varepsilon) \geq 1 - \varepsilon$ and $f\mathbf{1}_{K_\varepsilon}$ is continuous on K_ε.

Proof. □

6. Let $f \in L^1([0, 2\pi])$.

(i) Prove that $\lim_{n\to\infty} \int_0^{2\pi} f(x)\, e^{-inx}\, dx = 0.$

Proof. First observe that $|e^{-inx}| \leq 1$. By theorem 1.117 let g be a step function such that $\int_0^{2\pi} |f - g| < \epsilon$. Then by the triangle inequality,

$$|\int_0^{2\pi} fe^{-inx}dx| \leq |\int_0^{2\pi} (f-g)e^{-inx}dx| + |\int_0^{2\pi} ge^{-inx}dx|.$$

Since g is a step function given sufficiently large n we know $|\int_0^{2\pi} ge^{-inx}| \to 0$. For the other piece, notice

$$|\int_0^{2\pi} (f-g)e^{-inx}dx| \leq \int |f-g|dx < \epsilon.$$

Thus by substituting, $\int_0^{2\pi} f(x)^{e-inx}dx \leq \lim |\int_0^{2\pi} f(x)e^{-inx}dx| < \epsilon$ for sufficiently large n so $\lim_{n\to\infty} fe^{-inx}dx = 0$. □

(ii) Prove that for a set $E \subset [0, 2\pi]$ of positive Lebesgue measure $m(E)$,

$$\lim_{n\to\infty} \int_E \cos^2(nx + \gamma_n)\, dx = \frac{m(E)}{2},$$

for any sequence γ_n.

Proof. To begin use the trig identity for $cos^2(\theta)$ such that $cos^2(nx + \gamma_n) = \frac{1+cos(2nx+2\gamma_n)}{2}$. Then $\int_E cos^2(nx+\gamma_n)dx = \frac{m(E)}{2} + \frac{1}{2}\int_E cos(2nx+2\gamma_n)dx$. Using euler's formula notice $cos(2nx+2\gamma_n) = Re(e^{2nx}e^{2_n})$ so by substitution $\int_E cos(2nx + 2\gamma_n)dx = Re(e^{2\gamma_n}\int_E e^{2nx}dx) \leq |\int e^{2inx}dx|$ since $|e^{2i\gamma_n}| = 1$. By part i),

$$\lim_{n\to} \int_E e^{2inx}dx = 0$$

so

$$\int_E cos(2nx + 2\gamma_n)dx \to 0$$

since $E \subset [0, 2\pi]$. By substitution, $\lim_{n\to} \int_E cos^2(nx + \gamma_n)dx = \frac{m(E)}{2}$. □

(iii) Let $F_n(x) = a_n \cos nx + b_n \sin nx$. Assume that the series $\sum_{n=0}^{\infty} F_n(x)$ converges on a set E of positive measure. Show that $\lim_{n\to\infty} a_n = 0$ and $\lim_{n\to\infty} b_n = 0$. *Hint: It might help to express $F_n(x)^2$ in the form $(r_n \cos(nx + \gamma_n))^2$, with $r_n > 0$.*

Proof. Let $F_n(x)^2 = r^2 cos(nx + \gamma_n)^2$ where $r^2 = a_n^2 + b_n^2$. We want to show $r_n \to 0$. Since $\sum_{n=0}^{\infty} F_n(x)$ converges on E then for all $x \in E$, $F_n(x) \to 0$ and furthermore $F_n(x)^2 \to 0$ pointwise also. Using Egorov's theorem, there exists a subset $A \subset E$ such that $F_n^2(x) \to 0$ uniformly on A which shows $\int_A F_n^2(x)dx \to 0$. Since $\int_A F_n(x)^2 dx = r_n^2 \int_A cos(nx + \gamma_n)^2 dx$ then

$$r_n^2 = \frac{\int_A F_n(x)^2 dx}{\int_A cos(nx + \gamma_n)^2 dx}.$$

Using the result part ii)

$$r_n^2 \to \frac{0}{m(A)/2} = 0$$

thus $a_n^2 + b_n^2 \to 0$ so individually $a_n \to 0$ and $b_n \to 0$. □